14 Transition Metals: An Introduction

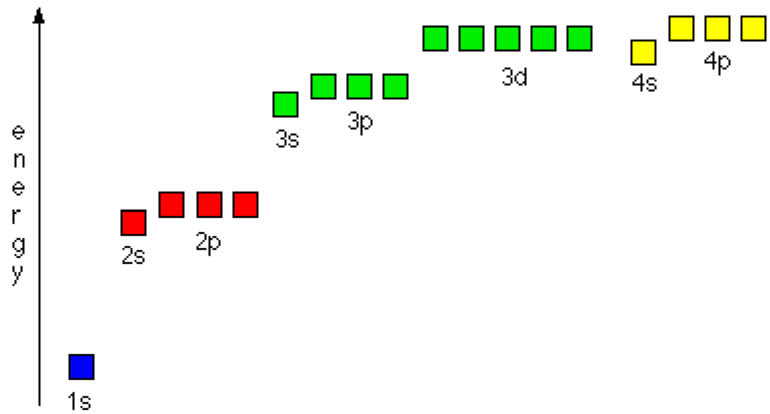
14.1 General Physical Properties

- A **transition element** is a metal which forms at least one stable ion with partially filled d-orbitals.
- A **d-block element** is an element which has at least one s-electron and at least one d-electron but no p-electrons in its outer shell.
- Not all d-block elements are transition elements: zinc and scandium are d-block elements but are not transition elements.
- The first row d-block elements which are also transition metals are therefore **titanium, vanadium, chromium, manganese, iron, cobalt, nickel and copper:**

1	2											13	14	15	16	17	18
				Key			1 H hydrogen 1.0										2 He helium 4.0
3 Li lithium 6.9	4 Be beryllium 9.0			atomic number atomic symbol name relative atomic mass								5 B boron 10.8	6 C carbon 12.0	7 N nitrogen 14.0	8 O oxygen 16.0	9 F fluorine 19.0	10 Ne neon 20.2
11 Na sodium 23.0	12 Mg magnesium 24.3	3	4	5	6	7	8	9	10	11	12	13 Al aluminium 27.0	14 Si silicon 28.1	15 P phosphorus 31.0	16 S sulfur 32.1	17 Cl chlorine 35.5	18 Ar argon 39.9
19 K potassium 39.1	20 Ca calcium 40.1	21 Sc scandium 45.0	22 Ti titanium 47.9	23 V vanadium 50.9	24 Cr chromium 52.0	25 Mn manganese 54.9	26 Fe iron 55.8	27 Co cobalt 58.9	28 Ni nickel 58.7	29 Cu copper 63.5	30 Zn zinc 65.4	31 Ga gallium 69.7	32 Ge germanium 72.6	33 As arsenic 74.9	34 Se selenium 79.0	35 Br bromine 79.9	36 Kr krypton 83.8

Electronic Configuration

- The outer electronic configuration of elements in the first row of the d-block is described below:



- **Electrons occupy the lower orbitals first. Thus the 4s fills before the 3d.**

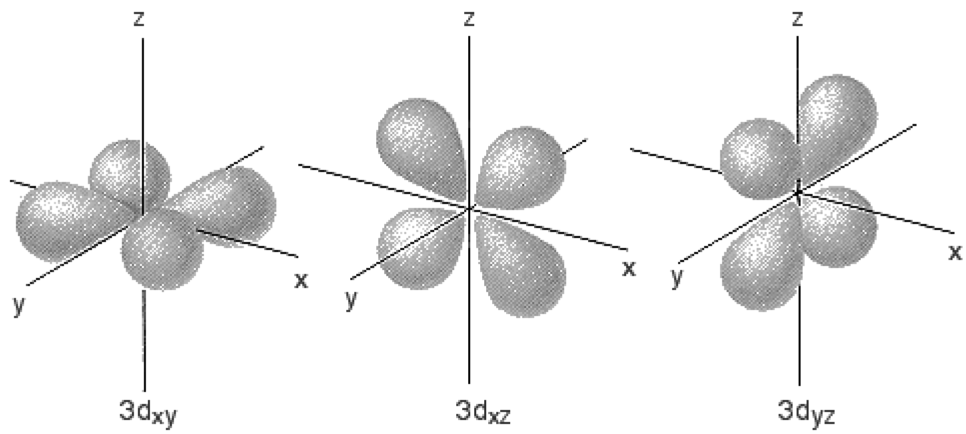
		4s	3d				
Sc	[Ar]	↑↓	↑				
Ti	[Ar]	↑↓	↑	↑			
V	[Ar]	↑↓	↑	↑	↑		
Cr	[Ar]	↑	↑	↑	↑	↑	↑
Mn	[Ar]	↑↓	↑	↑	↑	↑	↑
Fe	[Ar]	↑↓	↑↓	↑	↑	↑	↑
Co	[Ar]	↑↓	↑↓	↑↓	↑	↑	↑
Ni	[Ar]	↑↓	↑↓	↑↓	↑↓	↑	↑
Cu	[Ar]	↑	↑↓	↑↓	↑↓	↑↓	↑↓
Zn	[Ar]	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓

- Note the unusual structures of chromium and copper. The 4s and 3d subshells are closely similar in energy and therefore it is easy to promote electrons from the 4s into the 3d orbitals.
- In chromium the $4s^1 3d^5$ structure is adopted because the repulsion between two paired electrons in the 4s orbital is more than the energy difference between the 4s and 3d subshells. It is thus more stable to have unpaired electrons in the higher energy 3d orbital than paired electrons in the lower energy 4s orbital. Thus copper adopts a $4s^1 3d^{10}$ configuration.
- Electrons are removed from the outer shell first, so that **the 4s electrons are always removed before the 3d. The 3d electrons are only removed after all 4s electrons have been removed.**
  Some examples of electronic configurations of transition metal ions are shown below:

		4s	3d				
V^{2+}	[Ar]		↑	↑	↑		
Cr^{3+}	[Ar]		↑	↑	↑		
Mn^{2+}	[Ar]		↑	↑	↑	↑	↑

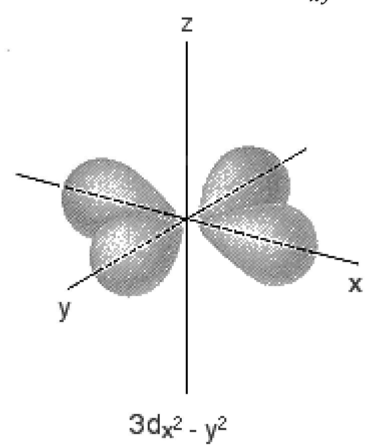
Shapes of d-Orbitals

- There are five 3d orbitals and these can be classified into two sets:
  (i) $\mathbf{3d_{xy}}$, $\mathbf{3d_{xz}}$ and $\mathbf{3d_{yz}}$ orbitals
  - These lie in the $x-y$ plane, the $x-z$ plane and the $y-z$ plane, respectively.
  - Each orbital has four lobes, with each lobe pointing **between** the stated two axes:

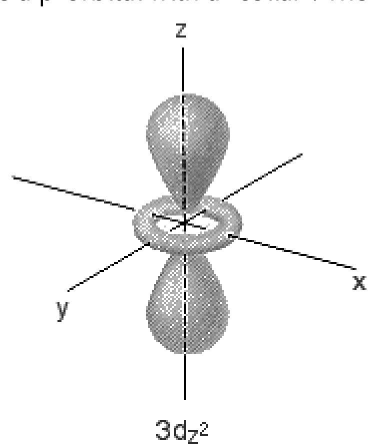


(ii) $\boldsymbol{3d_{z^2}}$ and $\boldsymbol{3d_{x^2-y^2}}$ orbitals

- the lobes of these orbitals point **along** the stated axes. For example, the lobes of the $3d_{x^2-y^2}$ point along the x and y axes, unlike those of the $3d_{xy}$ which point between the x and y axes.



- the $3d_{z^2}$ orbitals look like a p-orbital with a "collar". The main lobes point along the z –axis:



General Physical Properties of Transition Elements

(i) Variable Oxidation States

- The close proximity of 3d and 4s orbitals means that similar energies required for removal of 4s and 3d electrons.

- Maximum oxidation state rises across the group to manganese, and falls as the energy required to remove more electrons becomes very high.

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
				$+7$					
			$+6$	$+6$	$+6$				
		$+5$	$+5$	$+5$	$+5$				
	$+4$	$+4$	$+4$	$+4$	$+4$	$+4$			
$+3$	$+3$	$+3$	$+3$	$+3$	$+3$	$+3$	$+3$	$+3$	
	$+2$	$+2$	$+2$	$+2$	$+2$	$+2$	$+2$	$+2$	$+2$
								$+1$	

- Note that scandium only forms $+3$ ions (d^0). This is because the low effective nuclear charge on scandium enables all three valence electrons to be removed fairly easily.
- Note also that zinc only forms +2 ions (d^{10}). This is because the high effective nuclear charge on zinc prevents any 3d electrons from being removed.
  Thus scandium and zinc are not transition elements.

(ii) Higher Densities and Melting Points than s-block Elements

- Transition elements are all metals: the strong metallic bonds result from small atomic size and close packing of the atoms account for the high melting points.
- They **higher densities** than s-block metals of the same Period, such as Ca, because:
  1. They have smaller atomic size than the s-block elements (atomic radii decrease across the Periods);
  2. The atomic masses increase across the period.

$$density = \frac{mass}{volume}$$

	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
$m.p\ /°C$	63	850	850	1400	1677	1615	1260	1530	1495	1452	1080	420
$density/gcm^{-3}$	0.86	1.55	3.00	4.50	6.10	7.20	7.40	7.90	8.90	8.95	8.96	7.13

(iii) Formation of Complex Ions

- A **complex ion** is an ion comprising one or more ligands attached to a central metal cation by means of a dative covalent bonds.
- A **ligand** is a molecule or anion which can use its lone pair(s) of electrons to form a dative covalent bond with a transition metal. Examples of ligands are H_2O, NH_3, Cl^-, OH^-, CN^-,$^-OOCCOO^-$, and $H_2NCH_2CH_2NH_2$.
- Cations which form complex ions must have two features:
  1. they must have a high charge density, and thus be able to attract electrons from ligands;
  2. they must have empty orbitals of low energy, so that they can accept the lone pair of electrons from the ligands.
- Cations of transition metals are small, have a high charge and have available empty 3d and 4s orbitals of low energy. Therefore they form complex ions readily.

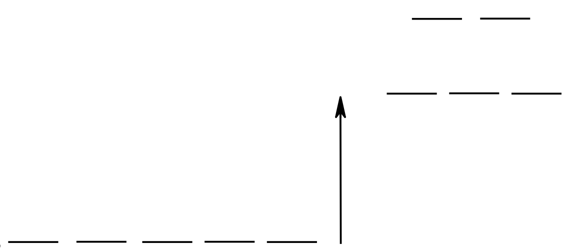
- The number of lone pairs of electrons which a cation can accept is known as the **coordination number** of the cation. It depends on the size and electronic configuration of that cation, and also on the size and charge of the ligand. The most common coordination number is 6, although 4 and 2 are also known.
- Examples of complex ions and their coordination numbers are given below. Note that the formula of the ion is always written inside square brackets with the overall charge written outside the brackets.

complex ion	$[\boldsymbol{Fe(H_2O)_6}]^{2+}$	$[\boldsymbol{CoCl_4}]^{2-}$	$[\boldsymbol{Cu(NH_3)_4(H_2O)_2}]^{2+}$
coordination number	6	4	6

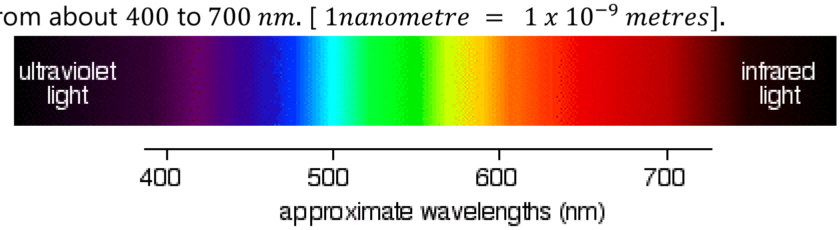
(iv) Formation of Coloured Complex Ions

- In an isolated atom or ion, the five 3-d orbitals of a transition metal are of the same energy. They are said to be **degenerate.**
- When a transition metal cation forms a complex ion, the incoming ligands repel the electrons in the atom, and thus they are raised in energy. Some of the d-orbitals, however, are repelled more than others and the result is that the d-orbitals are split into 2 groups of orbitals, with some orbitals being slightly lower in energy than the others: the orbitals are no longer degenerate.

degenerate d-orbitals



- The difference in energy between these groups of orbitals is similar to the energy of visible light. The energy of the light is related to the frequency of the light by the equation $E = hf$. Visible light has wavelength from about 400 to $700\ nm$. [$1 nanometre = 1 \times 10^{-9}\ metres$].



- If these d-orbitals are partially filled, some of the electrons in the lower energy orbitals are excited into the higher energy orbitals, and in doing so absorb the light that corresponds to that frequency. The resultant light is deficient in the light of that frequency and thus appears coloured.
- Thus two key conditions must be satisfied if the ion is to be coloured:
  1. **there must be a splitting of the d-orbitals**. This only happens in the presence of ligands and thus only complex ions are coloured. Anhydrous ions do not have split d-orbitals and so cannot absorb light in the visible spectrum and are thus white.
     For example, anhydrous $CuSO_4$ is white but hydrated $CuSO_4.5H_2O$ is cyan (blue).

  2. **the d-orbitals must be partially filled**. If the d-orbitals are empty (for example, in Sc^{3+}, Mg^{2+}) then there are no electrons which can be excited into the higher energy d-orbitals and the ions will be colourless. If the d-orbitals are full (for example, in Cu^{+}, Zn^{2+}) then there are no empty orbitals into which the electrons can be excited and the ions will also be colourless.

(v) Catalysis

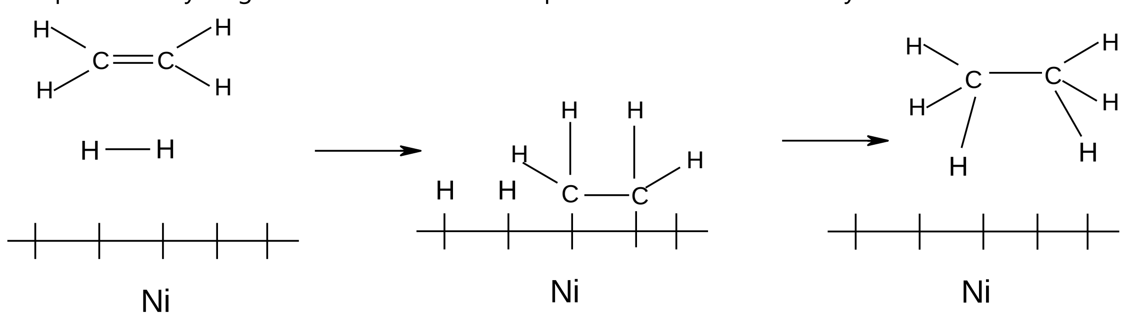
- The ability of transition metals to form more than one stable oxidation state enables them to catalyse certain redox reactions. They can be readily oxidised and reduced again, or reduced and then oxidised again, as a consequence of having a number of different oxidation states of similar stability. They can behave either as **homogeneous catalysts or as heterogeneous catalysts**.
- An example of **homogeneous catalysis** is the action of the Fe^{2+}/Fe^{3+} on the reaction between iodide ions and peroxodisulphate (VI) ions. Peroxodisulphate (VI) ions and iodide ions react together in aqueous solution to form sulphate (VI) ions and iodine:

$$\boldsymbol{S_2O_8^{2-}(aq) + 2I^-(aq) \xrightarrow{Fe^{2+}(aq)} 2SO_4^{2-}(aq) + I_2(aq)}$$

- If a small amount of Fe^{2+} ions are added to the reaction mixture, they will reduce the peroxodisulphate (VI) ions to sulphate (VI) ions and are themselves oxidised to Fe^{3+} ions in the process. The Fe^{3+} formed then oxidise iodide ions to iodine and are themselves reduced back to Fe^{2+} ions:

$$\boldsymbol{S_2O_8^{2-}(aq) + 2Fe^{2+}(aq) \longrightarrow 2SO_4^{2-}(aq) + 2Fe^{3+}(aq)} \quad \textbf{step 1}$$
$$\boldsymbol{2I^-(aq) + 2Fe^{3+}(aq) \longrightarrow I_2(aq) + 2Fe^{2+}(aq)} \quad \textbf{step 2}$$

- The cation reacts with each anion in turn, thus avoiding the need for a direct collision between two similarly charged ions (this is difficult since they repel each other).
- **Heterogeneous catalysts** react by allowing the reactant molecules to bond to the surface of the metal, usually by attracting the surface electrons. The reaction takes place at the surface, and the transition metal regains electrons as the products leave. The reaction is catalysed because the reactant molecules spend more time in contact with each other than they would in the absence of a catalyst.
An example is the hydrogenation of ethene in the presence of a Nickel catalyst:



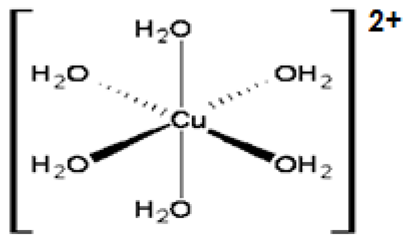
- Further examples of homogenous and heterogeneous catalysis involving transition metal ions or compounds are summarised below:

Reaction	Catalyst
$\boldsymbol{SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)}$	$V_2O_5(s)$
$\boldsymbol{2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)}$	$MnO_2(s)$
$\boldsymbol{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)}$	$Fe(s)$
$\boldsymbol{2NO(g) + 2CO(g) \longrightarrow N_2(g) + 2CO_2(g)}$	$Rh(s)$

14.2 General Chemical Properties

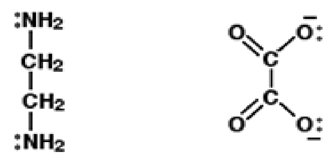
Types of Ligands

- Ligands such as H_2O, NH_3, Cl^-, OH^- and CN^- form only one dative covalent bond per ligand and are said to be **unidentate or monodentate.**

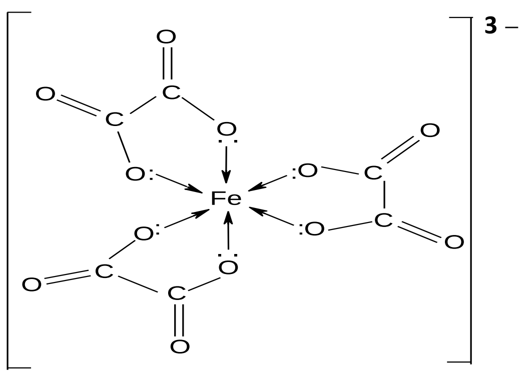


Dative bonding in the complex $[Cu(H_2O)_6]^{2+}$

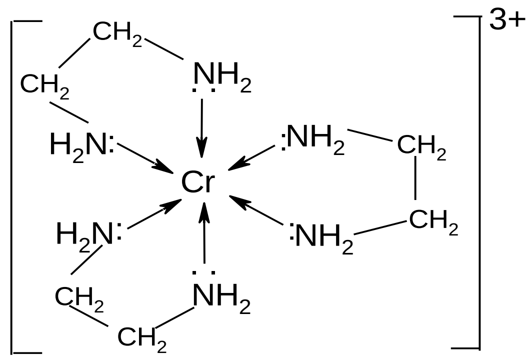
- Ligands such as the ethanedioate ions, $^-OOCCOO^-$, and 1,2-diaminoethane, $H_2NCH_2CH_2NH_2$ donate two lone pairs per ligand and are said to be **bidentate.**



1,2-diaminoethane   ethanedioate ion

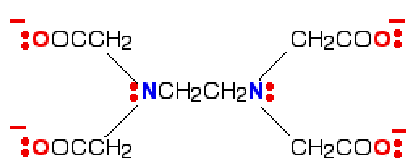


$[\boldsymbol{Fe(C_2O_4)_3}]^{3-}$



$[\boldsymbol{Cr(H_2NCH_2CH_2NH_2)_3}]^{3+}$

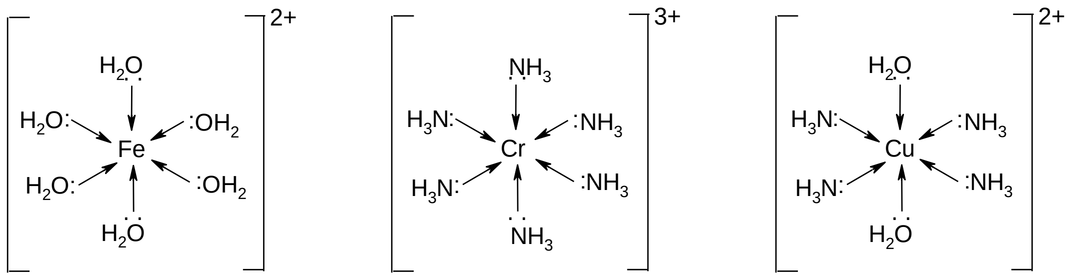
- The ligand $EDTA^{4-}$, or ethylenediaminetetraacetic acid, can form 6 dative covalent bonds per ligand. It is thus said to be **hexadentate.**



An example of a complex ion made from $EDTA^{4-}$ is $[\boldsymbol{Cu(EDTA)}]^{2-}$

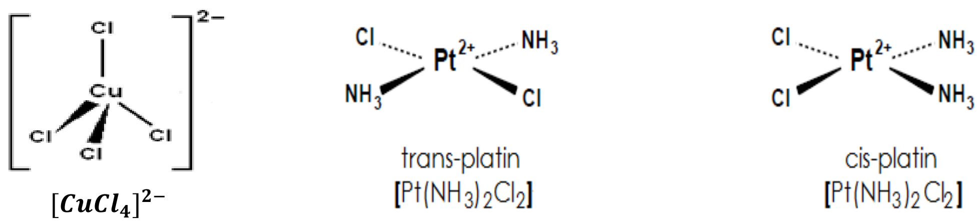
Shapes of Complex Ions

- **6-coordinate complexes made from monodentate ligands are all octahedral**. They can thus be drawn as follows:



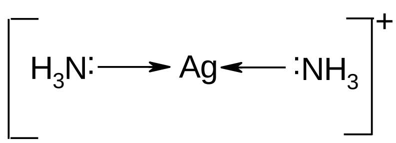
$[Fe(H_2O)_6]^{2+}$ $[Cr(NH_3)_6]^{3+}$ $[Cu(NH_3)_4(H_2O)_2]^{2+}$

- **4-coordinate complexes are generally tetrahedral but can also be square planar**, and are formed with larger ligands such as the chloride ion. Larger ligands cannot fit around the transition metal so easily and hence form the 4-coordination complexes.
  $[CuCl_4]^{2-}$ is tetrahedral while the the cisplatin and transplatin isomers are examples of square planar complexes.



$[CuCl_4]^{2-}$

- **2-coordinate complexes are generally linear**. They are mainly formed with Ag^+ ions. An example is the diamine silver(I) complex, drawn as follows:

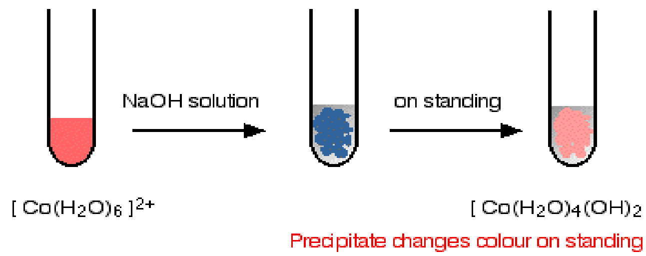
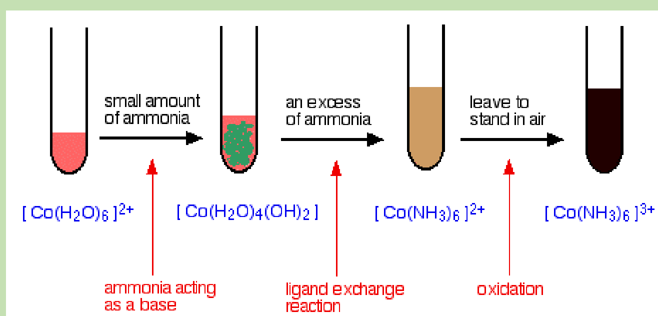
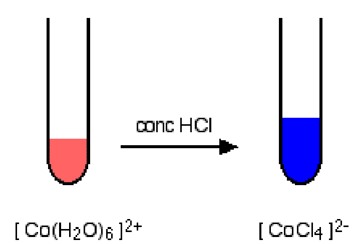


$[Ag(NH_3)_2]^+$

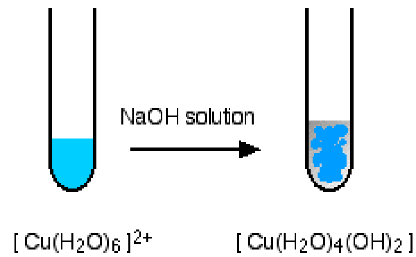
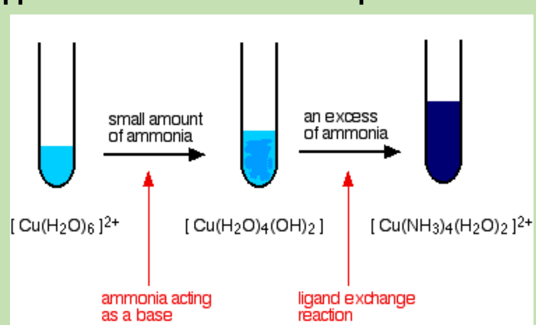
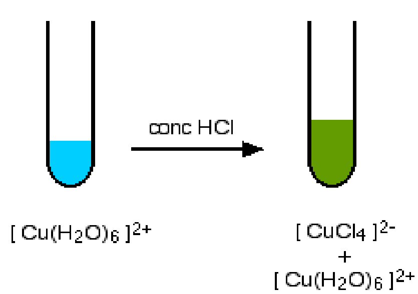
Ligand Exchange and Some Reactions of Transition Metal Ions

- **Ligand exchange** is the replacement of one ligand by another in a complex.
- With hexaaqua complexes, a little ammonia acts as a base and facilitates deprotonation. **Deprotonation** is the loss of a proton by a water ligand to form a hydroxo ligand. Deprotonation is caused by the high charge density on the central cation, which weakens the $O-H$ bonds in the water ligands, and enables the H^+ (the proton) to leave.
- Excess ammonia tends to cause ligand exchange to occur, replacing aqua and hydroxo ligands with ammine ligands.

(i) The Chemistry of Cobalt

Ligand	Chemistry
OH^-	$[Co(H_2O)_6]^{2+}(aq) + 2OH^-(aq) \longrightarrow [Co(OH)_2(H_2O)_4]\,(s) + 2H_2O(l)$ **pink, octahedral** **pink ppt. which changes to pink on standing soluble in xs. $NaOH$** 
NH_3	$[Co(H_2O)_6]^{2+}(aq) + 2NH_3(aq) \longrightarrow [Co(OH)_2(H_2O)_4]\,(s) + 2NH_4^+(aq)$ **pink, octahedral** **blue/pink ppt. soluble in xs. NH_3** The ammonia initially acts as a base. When added in excess, the ammonia acts as a ligand and displaces the water ligands: $[Co(OH)_2(H_2O)_4]\,(s) + 6NH_3(aq) \longrightarrow [Co(NH_3)_6]^{2+}(aq) + 4H_2O(l) + 2OH^-(aq)$ Ammonia makes cobalt (II) unstable, such that it is oxidised to cobalt (III) by air. $[Co(NH_3)_6]^{2+}(aq) \longrightarrow [Co(NH_3)_6]^{3+}(aq) + e^-$ **yellow, octahedral** **brown, octahedral** 
Cl^-	$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq) + 6H_2O(l)$ **pink, octahedral** **blue, tetrahedral** A reversible ligand displacement reaction occurs, forming a more stable, blue solution. A tetrahedral complex is more stable as there is less repulsion between ligands than in an octahedral complex. Adding excess water to the blue solution causes it to revert to pink. . This reaction is used in the test for water: blue cobalt chloride paper turns pink in the presence of water. 

(ii) The Chemistry of Copper

Ligand	Chemistry
OH^-	$[Cu(H_2O)_6]^{2+}(aq) + 2OH^-(aq) \longrightarrow [Cu(OH)_2(H_2O)_4]\,(s) + 2H_2O(l)$ **blue, octahedral** — **pale blue ppt.** **insoluble in xs.** $NaOH$ 
NH_3	$[Cu(H_2O)_6]^{2+}(aq) + 2NH_3(aq) \longrightarrow [Cu(OH)_2(H_2O)_4]\,(s) + 2NH_4^+(aq)$ **blue, octahedral** — **pale blue ppt.** **soluble in xs.** NH_3 The ammonia initially acts as a base. When added in excess, the ammonia acts as a ligand and displaces the water ligands: $[Cu(H_2O)_6]\,(aq) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}(aq)\ + 4H_2O(l)$ **pale blue ppt.** — **deep blue solution** 
Cl^-	$[Cu(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 6H_2O(l)$ **blue, octahedral** — **yellow, tetrahedral** A reversible ligand displacement reaction occurs, forming a more stable, yellow solution. The solution actually appears green due to the presence of the blue $[Cu(H_2O)_6]^{2+}(aq)$. Adding excess water to the yellow solution causes it to revert to blue. 

Redox Chemistry of Transition Elements

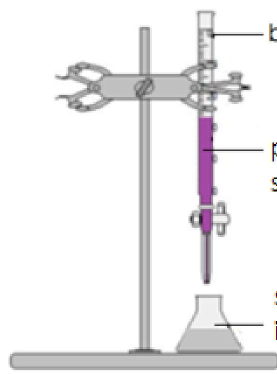
- Two important reagents involving transition metals that are used in redox titrations are potassium manganate (VII), $KMnO_4$, and potassium dichromate, $K_2Cr_2O_7$.

(i) Potassium Manganate (VII) ($KMnO_4$) Redox Reactions

- The MnO_4^- ion is a powerful oxidising agent which can be used to determine the amounts of reducing agents such as Fe^{2+}:

$$\underset{\text{intense purple}}{\boldsymbol{MnO_4^-(aq) + 8H^+(aq) + 5e^-}} \longrightarrow \underset{\text{colourless}}{\boldsymbol{Mn^{2+}(aq) + 4H_2O(l)}} \qquad \boldsymbol{E^\theta = +1.52\ V}$$

- The colour change is so intense that **no indicator is required** for the reaction.
- The $KMnO_4$ solution is generally placed in the burette. This causes two problems:
  (i) The intense colour of $KMnO_4$ makes difficult to see the graduation marks on the burette, and so it is difficult to make very accurate readings;
  (ii) The $KMnO_4$ reacts slightly with the glass, causing a slight stain if the burette is used too often.



- The $KMnO_4$ is placed in the burette instead of in the conical flask for the following reason:
  If the $KMnO_4$ were in the conical flask, it would slowly decolourise until the solution became completely colourless. The gradual disappearance of the pink colour is much harder to detect than the sudden appearance of the pink colour, and for this reason the $KMnO_4$ solution is always placed in the burette, despite the difficulties it presents in reading the burette.

- The reaction

$$\boldsymbol{MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l) \qquad E^\theta = +1.52\ V}$$

shows that the MnO_4^- ion is only an effective oxidising agent in acidic medium, as the presence of H^+ ions make the reaction go forward.

- Hence some acid is added to the conical flask before carrying out the titration. Sulphuric acid is generally used for this purpose. Hydrochloric acid should not be used as the MnO_4^- ends up reducing both the chloride ions in HCl and the reducing agent under investigation:

$$\boldsymbol{MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l) \qquad E^\theta = +1.52\ V}$$

$$\boldsymbol{Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq) \qquad E^\theta = +1.36\ V}$$

Overall: $\boldsymbol{2MnO_4^-(aq) + 16H^+(aq) + 10Cl^-(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5Cl_2(g)}$

- Common examples of reducing agents which are determined using $KMnO_4$ titrations are hydrogen peroxide solution, $Fe^{2+}(aq)$ ions and the ethanedioate ions, $C_2O_4^{2-}(aq)$, which has the structural formula $^-OOC-COO^-$:

$$\boldsymbol{Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-}$$

$$\boldsymbol{C_2O_4^{2-}(aq) \longrightarrow 2CO_2(g) + 2e^-}$$

$$\boldsymbol{H_2O_2(aq) \longrightarrow O_2(g) + 2H^+(aq) + 2e^-}$$

- Overall equations are as follows:

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq)$$
$$2MnO_4^-(aq) + 16H^+(aq) + 5C_2O_4^{2-}(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 10CO_2(g)$$
$$2MnO_4^-(aq) + 6H^+(aq) + 5H_2O_2(aq)(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5O_2(g)$$

Thus MnO_4^- reacts in a $1:5$ ratio with Fe^{2+} ions, and in a $2:5$ ratio with ethanedioate ions and with hydrogen peroxide.

- However, the reaction between ethanedioate ions and potassium manganate (VII) is carried out above $60°C$ due to the slow rate of reaction.

(ii) Potassium Dichromate (VI) ($K_2Cr_2O_7$) Redox Reactions

- The $Cr_2O_7^{2-}$ ion is a powerful oxidising agent which can be used to determine the amounts of reducing agents such as Fe^{2+}:

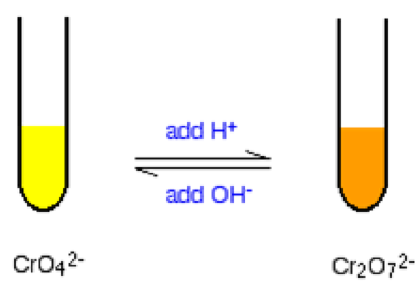
$$\underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + 14H^+(aq) + 6e^- \longrightarrow \underset{\text{green}}{2Cr^{3+}(aq)} + 7H_2O(l) \qquad E^\theta = +1.33\,V$$

- The colour change is not as intense as with the manganate titration and an indicator is needed. Diphenylaminesulphonate is used. It turns from colourless to purple at the end-point. Still, the end-point of a dichromate titration is difficult to see.
- The dichromate is always placed in the burette as it is easier to see the appearance of the purple colour than its disappearance.
- As with manganate, the dichromate ion is a good oxidising agent in acidic medium. Care must be taken as to which acid to use: sulphuric acid should be used as chloride ions in HCl could interfere with the titration, especially if in high concentration. Notice that chloride ions cannot be oxidised by dichromate ions under normal conditions, but can themselves oxidise the dichromate. (Compare the E^θ values).
- Solutions of chromium (VI) exist as either the orange dichromate ion, $Cr_2O_7^{2-}$, which is stable in acidic solution, or the yellow chromate ion, CrO_4^{2-}, which is stable in alkaline solution. The two can interconvert in a non-redox reaction:

$$\underset{\text{yellow}}{2CrO_4^{2-}(aq) + 2H^+(aq)} \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq) + H_2O(l)}$$

or:

$$\underset{\text{yellow}}{2CrO_4^{2-}(aq) + H_2O(l)} \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq) + 2OH^-(aq)}$$



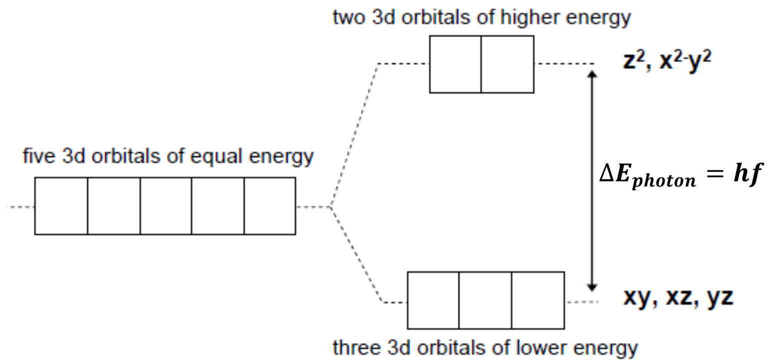
14.3 Colour of Complexes

The Origin of Colour

- The d-orbitals of a free transition metal atom or ion are degenerate (all have the same energy.) As the ligands approach, their electrons cause the d orbital electrons to be repelled unequally due to the orientation of the individual d orbitals.

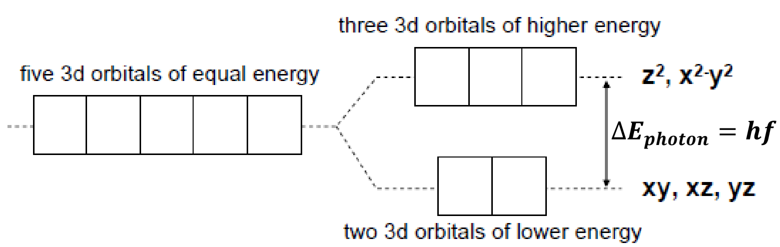
(i) d-Orbital Splitting in Octahedral Complexes

- For octahedral complexes, the ligands approach along the x, y, and z axes. Thus, the ligand electrons will repel the d orbitals that lie ON those axes, which are the $3d_{x^2-y^2}$ and $3d_{z^2}$ orbitals. These orbitals will experience stronger repulsions and thus will be at a higher energy state than the $3d_{xy}$, $3d_{xz}$ and $3d_{yz}$ orbitals which lie in-between the axes.



(ii) d-Orbital Splitting in Tetrahedral Complexes

- For tetrahedral complexes, the ligands will have a bit more interaction/repulsion with the orbitals in the planes of the axes – and thus the $3d_{xy}$, $3d_{xz}$ and $3d_{yz}$ orbitals will be the higher energy orbitals. This is opposite the octahedral situation.

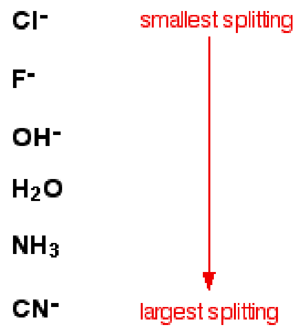


- The amount of splitting also depends on:
  (i) the nature of the ligands;
  (ii) oxidation state of the central metal atom/ion;
  (iii) the shape of the ligand.

The Nature of the Ligands

- Some ligands only produce a small energy separation among the d-orbitals while others cause a wider energy gap. The difference in energy between the new levels affects how much energy will be absorbed when an electron is promoted to a higher level; and hence what colour will be observed.

- The greater the splitting, the greater the energy required and the shorter the wavelength of the light absorbed. This means that as the splitting increases, the light absorbed tends to shift from red (long wavelength) towards orange-yellow, the shorter wavelength region.



- Thus, adding an excess of ammonia solution to hexaaquacopper (II) ions in solution, the pale blue (cyan) colour is replaced by a dark blue colour as the water molecules are replaced by the ammonia ligands of greater splitting.

The Oxidation State of the Metal

- As the oxidation state of the metal increase, so does the amount of splitting of the d-orbitals. Changes in oxidation state therefore affect the colour of the complex ion. For example:

Change of oxidation state results in change of colour

$[Cr(H_2O)_6]^{2+}$ $[Cr(H_2O)_6]^{3+}$

The Coordination of the Complex Ion

- Splitting is greater if the ion is octahedral than if it is tetrahedral. Therefore a change of coordination results in a change of colour.

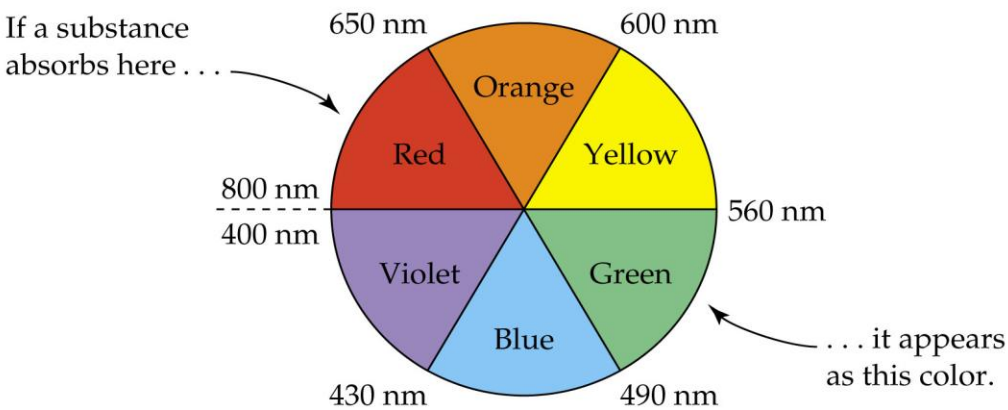
Change of coordination number results in change of colour. However, in this case, there has also been a change of ligand.

$[Co(H_2O)_6]^{2+}$ $[CoCl_4]^{2-}$

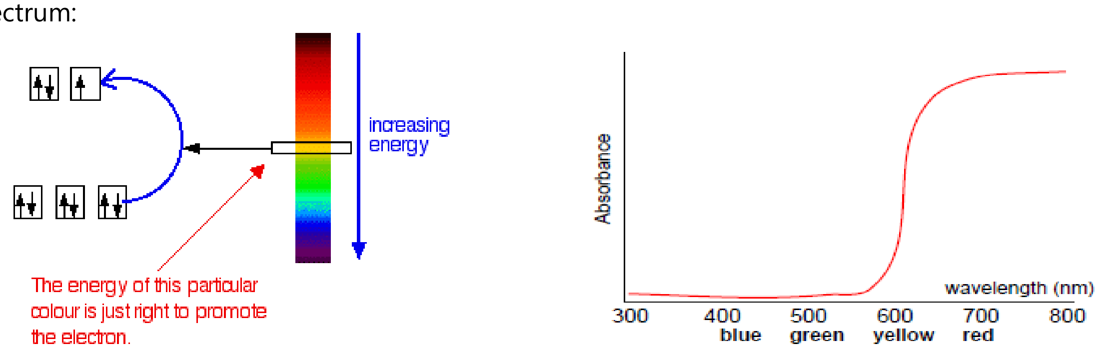
What Colour is Observed?

- When the d-level is not completely filled, electrons can be promoted from a lower energy d-orbital to a higher energy d-orbital by absorption of a photon of electromagnetic radiation having an appropriate energy $\Delta E_{photon} = hf$. The energy absorbed is within the visible region of the electromagnetic spectrum.

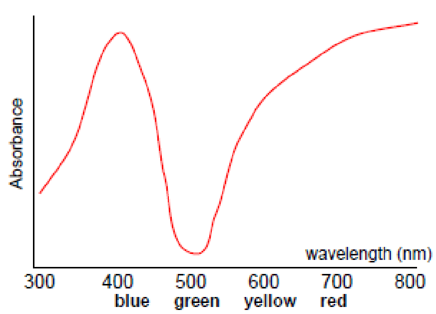
- When the wavelengths of one or more colours is absorbed, it is the complementary colour that is observed.
- Grass and leaves appear green because chlorophyll absorbs wavelengths in the red and violet portion of the visible spectrum. Solutions which absorb blue wavelengths will appear as orange, etc.



- Hence, solutions of $[Cu(H_2O)_6]^{2+}(aq)$ are blue because they absorb in the orange-red region of the spectrum:



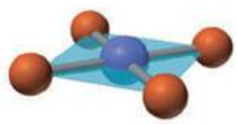
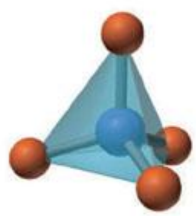
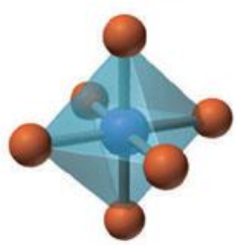
- The following complex should be green in colour, as that is the least absorbed:



14.4 Stereoisomerism

Shapes of Complexes

- The geometry or shape of a complex ion depends on the coordination number and the nature of the metal ion. If there are 2 ligands attached to a central metal ion, then the geometry is linear, if there are 4 ligands, then square planar or tetrahedral, and 6 will result in octahedral geometry.
- The difference between the square planar and tetrahedral geometries depends on the central metal ion. Most d^8 ions are square planar while most d^{10} ions are tetrahedral.

Coordination Number	Shape	Structure	Examples
2	Linear		$[CuCl_2]^-$ $[Ag(NH_3)_2]^+$
4	Square planar		$[Ni(CN)_4]^{2-}$ $[Pt(NH_3)_4]^{2+}$
4	Tetrahedral		$[CoCl_4]^{2-}$ $[Cu(CN)_4]^{2-}$
6	Octahedral		$[FeCl_6]^{3-}$ $[Co(en)_3]^{3+}$ $[Cu(NH_3)_4(H_2O)_2]^{2+}$

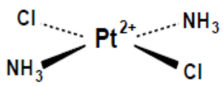
Isomerism in Complexes

- Isomers are two or more different compounds which have the same formula. There are two main forms of isomerism:

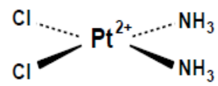
Stereoisomerism	Structural Isomerism
1. Geometrical (cis-trans) isomerism 2. Optical isomerism	1. Ionisation isomerism 2. Hydrate isomerism

(i) Stereoisomerism

- Stereoisomers have the same atoms, same sets of bonds, but differ in the orientation of these bonds in space.
- **Geometric isomerism** is possible for both square planar and octahedral complexes, but not tetrahedral. Example of molecule showing cis-trans isomerism is diamine-dichloro-platin (II), $[Pt(NH_3)_2Cl_2]$, commonly known as platin:

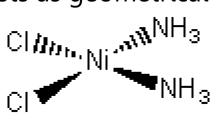


trans-platin
$[Pt(NH_3)_2Cl_2]$

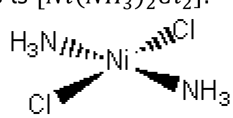


cis-platin
$[Pt(NH_3)_2Cl_2]$

- In the cis-isomer, the two ammonia ligands are adjacent to one another and the two chloro ligands are adjacent to one another. This compound is used medicinally to treat certain types of cancer.
- In the trans-isomer, the two ammonia ligands are opposite one another and the two chloro ligands are opposite one another.
- These are different compounds, as shown by the fact that the trans-isomer shows no anti-cancer while the cis-isomer is pharmacologically active as an anti-cancer drug.
- Another example of a square planar complex that exists as geometrical isomers is $[Ni(NH_3)_2Cl_2]$:



$\boldsymbol{cis} - [\boldsymbol{Ni(NH_3)_2Cl_2}]$



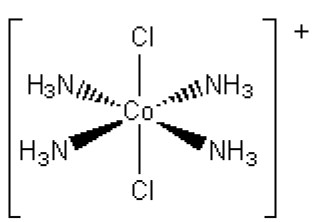
$\boldsymbol{trans} - [\boldsymbol{Ni(NH_3)_2Cl_2}]$

How cis-platin treats cancer

- *When cis-platin gets into the body, its neutral overall charge means that it can cross the cell membrane.*
- *Once in a cell it becomes activated by the replacement of one of the chlorides by a water molecule.*
- *The chloride falls off because the concentration of chloride within a cell is much less than it is in the bloodstream.*
- *The water itself is, in turn, easily displaced by the basic nitrogen atoms on DNA. Once bound to DNA the second chloride ion is replaced by a guanine nitrogen atom from an adjacent DNA strand.*
- *The result is a platinum fragment cross-linking two DNA strands within the double helix. This cross-linking prevents the cell dividing by mitosis and so the tumour stops growing.*
- *The damaged DNA can be repaired in healthy cells by DNA repair enzymes, but in tumour cells the 'kink' induced by the platinum cross-link is not recognised and the DNA cannot be fixed. As a result the cell undergoes programmed cell suicide - apoptosis - and the tumour shrinks.*

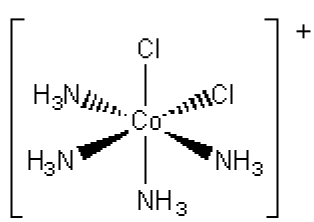
Source: Royal Society of Chemistry
http://www.rsc.org/chemistryworld/podcast/CIIEcompounds/transcripts/cisplatin.asp

- An example of a an octahedral complex exhibiting cis-trans isomerism is $[Co(NH_3)_4Cl_2]^+$.



$\boldsymbol{trans} - [\boldsymbol{Co(NH_3)_4Cl_2}]^+$

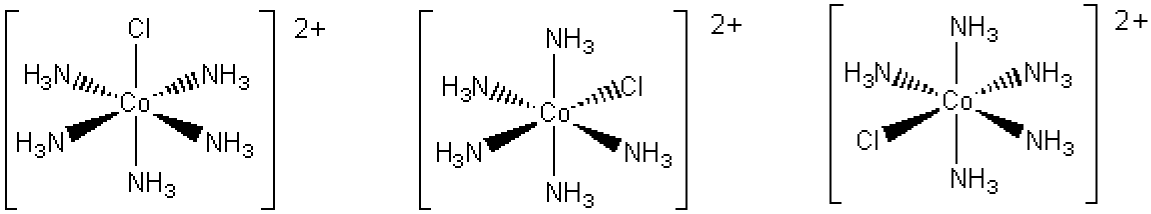
In the trans-isomer, the chloride ligands lie above and below the plane, in the axial positions



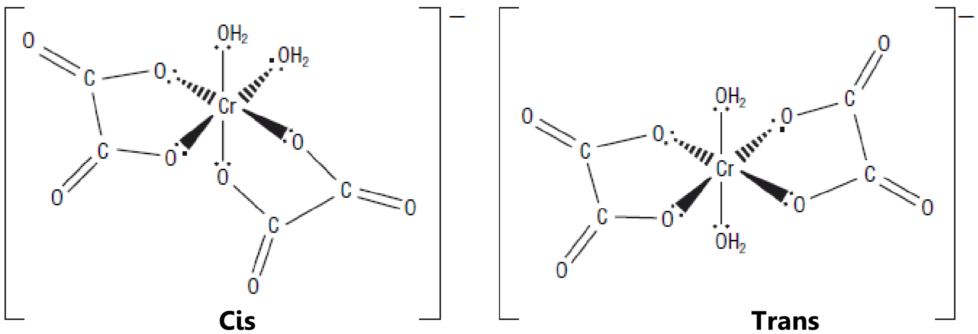
$\boldsymbol{cis} - [\boldsymbol{Co(NH_3)_4Cl_2}]^+$

In the cis-isomer, an ammonia molecule and a chloride ion lie in the axial positions.

- Note that an octahedral complex such as $[Co(NH_3)_5Cl]^{2+}$ does not have isomers. Hence the following structures all represent the same compound:

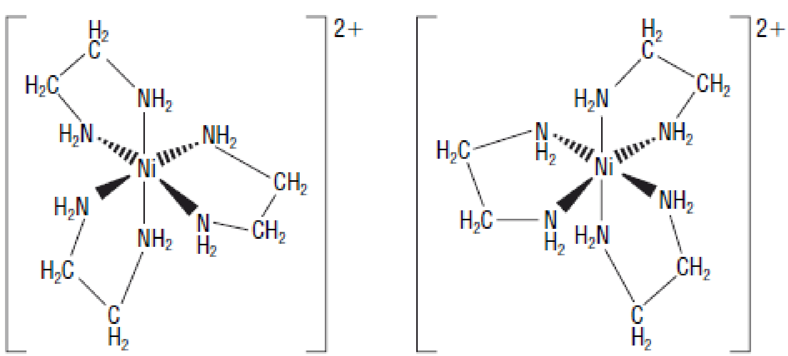


- The following is a case of bidentate ligands involved in cis-trans isomerism:



$[\boldsymbol{Cr(C_2O_4)_2(H_2O)_2}]^-$

- **Optical isomerism** is possible for both tetrahedral and octahedral complexes, but not square planar.
- Optical isomerism results in a pair of non-superimposable mirror images which rotate the plane of polarised light in opposing directions.
- A mixture containing equal amounts of the two isomers has no effect on plane-polarised light as the rotations cancel out.
- Optical isomerism arises when:
  (i) a complex has three molecules or ions of a bidentate ligand. An example is $[Ni(en)_3]^{3+}$;
  (ii) a complex has two molecules or ions of a bidentate ligand, and two molecules or ions of a monodentate ligand. An example is $[Ni(en)_2Cl_2]$
  (iii) a complex has one hexadentate ligand. An example is $[Cu(EDTA)]^{2-}$



$[\boldsymbol{Ni(H_2NCH_2CH_2NH_2)_3}]^{2+}$

The two are non-superimposable mirror images of each other.

(iv) Structural Isomerism

- **Ionisation isomerism** when compounds form different ions in solution. Example:

  $[PtBr(NH_3)_3].NO_2$ gives NO_2^- anions in solution. The bromide ions are not free to leave the complex;

  $[PtNO_2(NH_3)_3].Br$ gives Br^- anions in solution. The NO_2^- ions are not free to leave the complex.

- **Hydrate isomerism** is a form of ionisation isomerism, but with a water molecule playing the interchange role. For example, there are three different isomers of the compound $CrCl_3.H_2O$:

Isomer	Colour	On addition of $AgNO_3(aq)$
$[CrCl_2(H_2O)_4]Cl.2H_2O$	dark green	One mole of complex gives one mole of chloride ions
$[CrCl(H_2O)_5]Cl_2.H_2O$	grey-green	One mole of complex gives two moles of chloride ions
$[Cr(H_2O)_6]Cl_3$	violet	One mole of complex gives three moles of chloride ions

14.5 Stability Constants

Stability Constants, K_{stab}

- The stability constant, K_{stab}, of a complex ion is the equilibrium constant for the formation of the complex ion in a solvent from its constituent ions or molecules.
- Consider the equilibrium:

$$[Cu(H_2O)_6]\,(aq) + 4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4(H_2O)_2]^{2+}(aq) + 4H_2O(l)$$

Hence:

$$K_{stab} = \frac{[Cu(NH_3)_4(H_2O)_2]^{2+}}{[Cu(H_2O)_6]\,[NH_3]^4} \quad \textit{and the units are:} \quad \frac{mol\ dm^{-3}}{(mol\ dm^{-3})(mol\ dm^{-3})^4} = dm^{12}\ mol^{-4}$$

- The water has been left out of the K_{stab} expression since its concentration is virtually constant as everything is dissolved in the water.
- A large value of K_{stab} indicates that the position of an equilibrium lies far to the right. **Complex ions with high stability constants are more stable than those with lower values.** A large value of a stability constant shows that the ion is easily formed.
- For example, $[Co(NH_3)_6]^{2+}$ has a stability constant value of $1.35\ x\ 10^{23}\ dm^{18}\ mol^{-6}$ while $[Ni(NH_3)_6]^{2+}$ has a stability constant value of $8.98\ x\ 10^{8}\ dm^{18}\ mol^{-6}$, meaning that $[Co(NH_3)_6]^{2+}$ is more stable and more easily formed than $[Ni(NH_3)_6]^{2+}$.
- Bidentate and polydentate ligands, such as EDTA, form particularly stable complex ions. The additional stability of these complex ions results from an increase in entropy. Ligand substitution by EDTA is accompanied by a large increase in the total number of moles, which increases the entropy:

$$[Cu(H_2O)_6]\,(aq) + EDTA^{4-}(aq) \rightleftharpoons [Cu(EDTA)]^{2-}(aq) + 6H_2O(l) \quad K_{stab} = 6.30\ x\ 10^{18}\ dm^3\ mol^{-1}$$

- The left hand side has 2 moles, the right hand side has 7 moles; hence there is an increase in entropy which favours the forward reaction. The K_{stab} is very large.

$$[Cu(H_2O)_6]\,(aq) + 4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4(H_2O)_2]^{2+}(aq) + 4H_2O(l) \quad K_{stab} = 1.20\ x\ 10^{13}\ dm^{12}\ mol^{-4}$$

- Either side has five moles, hence there is no change in entropy. The stability constant is much less than when substituting the hexadentate ligand $EDTA^{4-}$.

Examination Practice 13
FREE RESPONSE QUESTIONS

1. Iron forms several complex ions in which the oxidation state of iron is +3.

One of these complex ions is $[Fe(CN)_6]^{3-}$. This is called the hexacyanoferrate (III) ion. In the hexacyanoferrate (III) ion the cyanide ions, CN^-, act as ligands.

(i) Suggest why a cyanide ion can act as a ligand.

(ii) Draw the expected shape for the complex ion $[Fe(CN)_6]^{3-}$. Include bond angles and the name of the shape.

[1 + 2]
[Total 3 marks]

2. Cobalt readily forms complex ions in which the cobalt has an oxidation state of +2.

One complex ion of cobalt is the hexaaquacobalt (II) ion $[Co(H_2O)_6]^{2+}$.

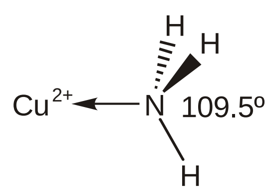
(i) What is the co-ordination number of Co^{2+} in this complex ion?

(ii) Water is acting as a ligand. Explain the meaning of the term *ligand*.

[1 + 2]
[Total 3 marks

3. Ammonia is a simple molecule. The H—N—H bond angle in an isolated ammonia molecule is 107°.

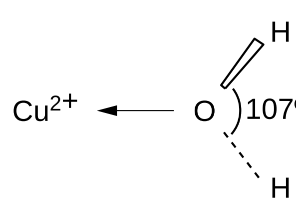
The diagram shows part of the $[Cu(NH_3)_4(H_2O)_2]^{2+}$ ion and the H—N—H bond angle in the ammonia ligand.



Explain why the H—N—H bond angle in the ammonia ligand is 109.5° rather than 107°.

[Total 3 marks]

4. Water is a simple molecule. The H—O—H bond angle in an isolated water molecule is 104.5°. The diagram shows part of the $[Cu(H_2O)_6]^{2+}$ ion and the H—O—H bond angle in the water ligand.



Explain why the H—O—H bond angle in the water ligand is 107° rather than 104.5°.

[Total 3 marks]

5. Artists between the 13th and the 19th Centuries used a green pigment called *verdigris.* The artists made the pigment by hanging copper foil over boiling vinegar.

A sample of *verdigris* has the formula $[(CH_3COO)_2Cu]_2.Cu(OH)_2.xH_2O$. Analysis of the sample shows that it contains 16.3% water by mass. Calculate the value of $\boldsymbol{x}$ in the formula.

[Total 3 marks]

6. In this question, one mark is available for the quality of use and organisation of scientific terms. Copper and iron are typical transition elements. One of the characteristic properties of a transition element is that it can form complex ions.

- Explain in terms of electronic configuration why copper is a transition element.
- Give an example of a complex ion that contains copper. Draw the three dimensional shape of the ion and describe the bonding within this complex ion.
- Transition elements show typical metallic properties. Describe **three other** typical properties of transition elements. Illustrate each property using copper or iron or their compounds.

[11]

Quality of Written Communication [1]

[Total 12 marks]

7. Dilute aqueous copper (II) sulphate contains $[Cu(H_2O)_6]^{2+}$ ions.

(a) Concentrated hydrochloric acid is added drop by drop to a small volume of dilute aqueous copper (II) sulphate. The equation for the reaction taking place is as follows.

$$[Cu(H_2O)_6]^{2+}(aq) + 4Cl^{-}(aq) \longrightarrow [CuCl_4]^{2-}(aq) + 6H_2O(l)$$

(i) Describe the observations that would be made during the addition of the concentrated hydrochloric acid.

(ii) Describe the bonding within the complex ion, $[CuCl_4]^{2-}$.

[1 + 2]

(b) Concentrated aqueous ammonia is added drop by drop to aqueous copper(II) sulphate until present in excess. Two reactions take place, one after the other, to produce the complex ion $[Cu(NH_3)_4(H_2O)_2]^{2+}(aq)$. Describe the observations that would be made during the addition of concentrated aqueous ammonia.

[2]

[Total 5 marks]

8. The Co^{2+} ion can form complexes with two different co-ordination numbers.

(a) What is meant by the *co-ordination number* of a complex ion?

[1]

(b) The following equilibrium is readily established.

$$[Co(H_2O)_6]^{2+} + 4Cl^{-} \rightleftharpoons [CoCl_4]^{2-} + 6H_2O$$

(i) Draw the shape of each complex ion in this equation.

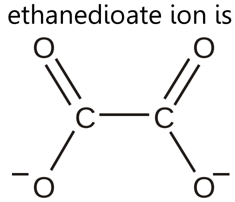
(ii) What colour change would you expect to see when an excess of Cl^{-} is added to $[Co(H_2O)_6]^{2+}$?

(iii) Describe how you would move the position of this equilibrium to the left.

[2 + 2 + 1]

[Total 6 marks]

9. The ethanedioate ion, $C_2O_4^{2-}$, can act as a bidentate ligand when it forms complex ions with a transition metal ion. The structure of the ethanedioate ion is shown below.



(a) What do you understand by the term *bidentate ligand?*

[2]

(b) The ethanedioate ion readily forms an octahedral complex ion with Cr^{3+}.
Show the structure and charge of this complex ion.

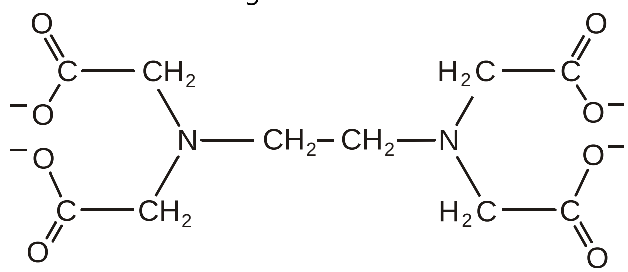
[3]
[Total 5 marks]

10. The $edta^{4-}$ ion forms complex ions with Ni^{2+}(aq).

(a) Write the electronic configuration of the Ni^{2+} ion.

[1]

(b) The $edta^{4-}$ ion has the following structure.



(i) Put a ring around two different types of atom in the $edta^{4-}$ ion that are capable of forming a dative covalent bond with the Ni^{2+} ion.

(ii) What feature of these atoms allows them to form a bond with Ni^{2+}?

[2 + 1]
[Total 4 marks]

11. Stereoisomerism is very common in transition metal complexes. Some complexes have found an important use in the treatment of cancer.

(i) Name a transition metal complex used in the treatment of cancer.

(ii) Describe how this complex helps in the treatment of cancer.

[1 + 2]
[Total 3 marks]

12. (a) Co^{2+} forms the complex $[Co(NH_3)_4Cl_2]$. This complex exists as two stereoisomers.

(i) Draw diagrams to show the two isomeric forms of this complex.

(ii) What type of stereoisomerism is shown by this complex?

[2 + 1]

(b) Cobalt also forms a complex with the formula $[Co(H_2NCH_2CH_2NH_2)_2Cl_2]$. This complex shows the same kind of isomerism as $[Co(NH_3)_4Cl_2]$ but it also shows a different type of stereoisomerism. Draw diagrams to show the two isomers of this different type of stereoisomerism.

[2]
[Total 5 marks]

13. (a) A common ligand which combines with a number of transition metal ions is ethane-1,2-diamine, $H_2NCH_2CH_2NH_2$. This is a bidentate ligand.
Explain the meaning of the term *bidentate*.

[1]

(b) The complex $[CoCl_2(H_2NCH_2CH_2NH_2)_2]$ is a neutral molecule. It shows two types of stereoisomerism. Use this molecule to explain what you understand by the term *stereoisomerism*. Your answer should include diagrams to show clearly the structures of the different isomers in both types of stereoisomerism.

[7]
[Total 8 marks]

14. Platinum forms complexes with a co-ordination number of 4.

(a) (i) Explain the term *co-ordination number*.

(ii) State the shape of these platinum complexes.

[2]

(b) The tetrachloroplatinate (II) ion readily undergoes the following reaction.

$$[PtCl_4]^x + 2NH_3 \rightleftharpoons [Pt(NH_3)_2Cl_2]^y + 2Cl^-$$

(i) What type of reaction is this?

(ii) Suggest values for x and y in the equation.

[2]

(c) The complex $[Pt(NH_3)_2Cl_2]^y$ exists in two isomeric forms.

(i) Draw diagrams to show the structure of these isomers.

(ii) What type of isomerism is this?

(iii) One of the isomers of $[Pt(NH_3)_2Cl_2]^y$ is an important drug used in the treatment of cancer. How does this drug help in the treatment of cancer?

[2 + 2 + 2]

[Total 10 marks]

15. This question looks at the chemistry of transition elements.

(a) (i) Explain what is meant by the terms *transition element*, *complex ion* and *ligand*,

(ii) Discuss, with examples, equations and observations, the typical reactions of transition elements.

In your answer you should make clear how any observations provide evidence for the type of reaction discussed.

[11]

(b) Describe, using suitable examples and diagrams, the different shapes and stereoisomerism shown by complex ions.

In your answer you should make clear how your diagrams illustrate the type of stereoisomerism involved.

[9]

[Total 20 marks]

16. The percentage by mass of iron in a sample of moss killer can be determined by titration against acidified potassium manganate (VII).

- Stage 1 – A sample of moss killer is dissolved in excess sulphuric acid.
- Stage 2 – Copper turnings are added to the acidified sample of moss killer and the mixture is boiled carefully for five minutes. Copper reduces any iron (III) ions in the sample to give iron(II) ions.
- Stage 3 – The reaction mixture is filtered into a conical flask to remove excess copper.
- Stage 4 – The contents of the flask are titrated against aqueous potassium manganate (VII).

(i) Suggest why it is important to remove all the copper in stage 3 before titrating in stage 4.

[1]

(ii) The ionic equation for the redox reaction between acidified MnO_4^- and Fe^{2+} is given below.

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq)$$

Explain, in terms of electron transfer, why this reaction involves both oxidation and reduction.

[2]

(iii) A student analyses a 0.675 g sample of moss killer using the method described.

In stage 4, the student uses 22.5 cm^3 of 0.0200 mol dm^{-3} MnO_4^- to reach the endpoint.

Calculate the percentage by mass of iron in the moss killer.

[4]

[Total 7 marks]

17. The transition element iron is the most abundant element in the Earth's core.

(a) What is meant by the term *transition element*?

(b) In aqueous solution, iron can form complex ions which contain ligands.

(i) Name the *type of bonding* that occurs between a ligand and a transition element.

[1]

(ii) Which of the following species can act as a ligand?

Copy and complete the table by placing a tick (✓) in the appropriate column to indicate whether the species can act as a ligand or not.

species	can act as a ligand	cannot act as a ligand
NO_3^-		
BF_3		
$H_2NCH_2CH_2NH_2$		
NH_4^+		

[1 + 2]

(c) Manganese ions, Mn^{2+}, show some similar chemical properties to those of copper (II) ions, $Cu^{2+}(aq)$. Use this information and the *Data Booklet* to suggest the formula of the manganese species formed in each of the following reactions. State the *type of reaction* taking place in each case.

	formula of manganese species formed	type of reaction
Mn^{2+}(aq) + NaOH(aq)		
Mn^{2+}(aq) + concentrated HCl		
Mn^{2+}(aq) + H_2O_2(aq)		

[5]

[Total 9 marks]

[© UCLES 2015 9701/42/O/N/15**]**

18. (a) Write the electronic configuration of chromium.

[1]

(b) Explain in terms of their electronic configurations why Fe^{2+} ions are readily oxidised to Fe^{3+} but Mn^{2+} ions are not readily oxidised to Mn^{3+} ions.

(c) Using iron and its compounds show what is meant by the terms

(i) homogeneous catalysis

(ii) heterogeneous catalysis

Include equations in your explanations.

[6]

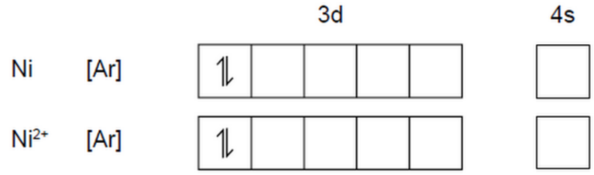
(d) With the aid of a balanced equations explain why aqueous solutions of iron (III) salts have a pH less than 5.

[3]

[Total 12 marks]

[© ZIMSEC 9189/1 J2007**]**

19. (a) Copy and complete the electron configurations for Ni and Ni^{2+}.

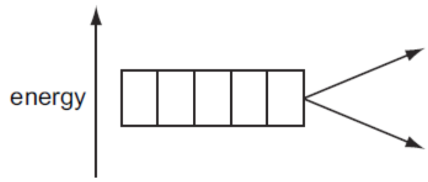


[2]

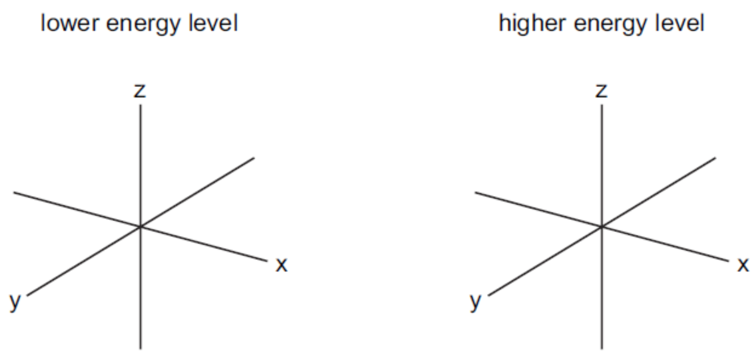
(b) The presence of electrons in d orbitals is responsible for the colours of transition element compounds.

(i) The d orbitals in an isolated transition metal atom or ion are all at the same energy level. What term is used to describe orbitals that are at the same energy level?

(ii) Complete the diagram to show the splitting of the d orbital energy levels in an octahedral complex ion.



(iii) On the axes below, sketch the shapes of one d orbital from the lower energy level and one d orbital from the higher energy level.



[3]

(c) The octahedral complex $[Ni(H_2O)_6]^{2+}$ is green. Explain the origin of the colour of this complex.

[4]

(d) When $NH_3(aq)$ is added to the green solution containing $[Ni(H_2O)_6]^{2+}$, a grey-green precipitate, **A**, is formed. This precipitate dissolves in an excess of $NH_3(aq)$ to give a blue-violet solution, **B**.
Suggest formulae for **A** and **B** and write equations for the two reactions producing **A** and **B**.

[4]

[Total 13 marks]

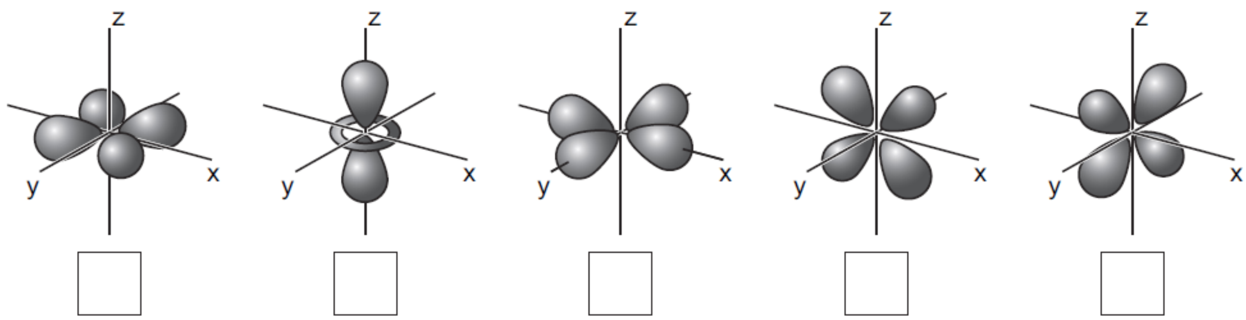
[© UCLES 2014 9701/42/O/N/14**]**

20. Transition elements have characteristic properties due to their partially-filled d orbitals.

(a) (i) Which two elements in the first row of the d-block have only one electron in the 4s orbital of their neutral atoms?

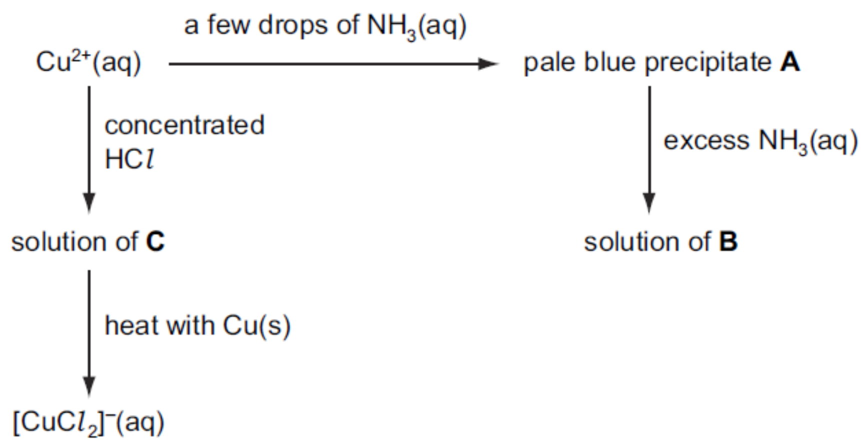
(ii) The d orbitals in an isolated transition metal atom or ion are described as being degenerate. What is meant by the term *degenerate*?

(iii) Sketches of the shapes of the atomic orbitals from the d subshell are shown.
In an octahedral complex, the d orbitals are split into two groups at different energy levels.
On the diagram below, write an 'H' in the box under each of the orbitals at the higher energy level.



[4]

(b) The following scheme shows some reactions of Cu^{2+}(aq).



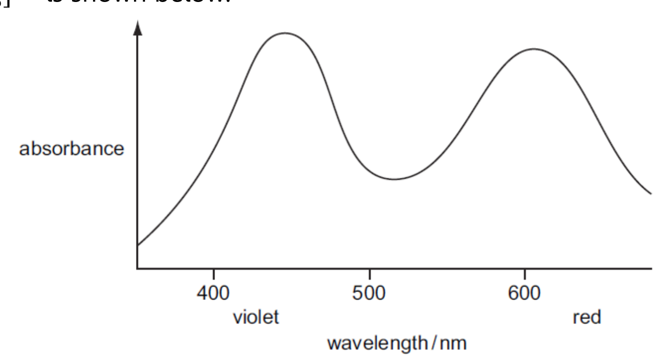
(i) Suggest the formula of each of A, B and C.
(ii) State the colour of the solutions B and C.
(iii) Name the type of reaction that occurs when C is heated with copper.
Deduce the role of the copper metal in this reaction.

[6]

(c) When the solution containing the complex $[CuCl_2]^-$ is poured into water, a precipitate of $CuCl$ is formed. $CuCl$ is white because it does not absorb visible light.
Explain why $CuCl$ does **not** absorb visible light.

[2]

(d) The complex ion $[Cr(H_2O)_6]^{3+}$ is coloured because it absorbs visible light. The absorption spectrum for $[Cr(H_2O)_6]^{3+}$ is shown below.



Suggest the colour of this complex ion. Explain your answer.

[2]
[Total 14 marks]

21. (a) (i) Define the term transition element.

(ii) State **three** general characteristics of a transition element or its compounds, illustrating your answer by examples from the chemistry of iron.

[4]

(b) Explain why an aqueous solution of iron (III) chloride has a pH less than 7.

[2]

(c) A steel chain of mass 10 g was dissolved in water acidified with dilute sulphuric acid and made up to 1 620 cm^3 of solution.

During the titration 20 cm^3 of the solution required 25 cm^3 of 0.03 $mol\ dm^{-3}$ $KMnO_4$.

(i) Give the colour change at the end point.

(ii) Calculate the percentage by mass of iron in the steel chain.

[5]

(d) State **one** biochemical importance of iron.

[1]

[Total 12 marks]

[© ZIMSEC 9189/1 N2011**]**

22. The following passage is taken from an A level Chemistry text book.

"In an isolated atom, the five d-orbitals have the same energy. In an octahedral complex ion, however, the presence of the ligands splits the five orbitals into a group of three and a group of two. These two groups have slightly different energies."

(a) Use the following sets of axes to draw the shape of **one** d-orbital in **each** of the two groups mentioned above.

[2]

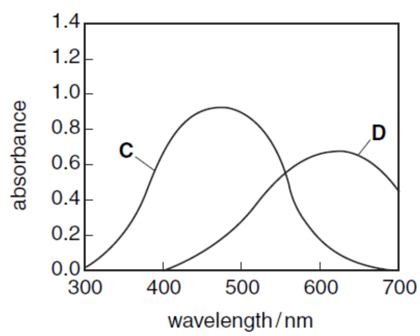
(b) Explain how the presence of the six ligands, L, in $[FeL_6]^{3+}$ splits the 3d orbitals into two groups of different energy, and explain whether the two-orbital group or the three-orbital group has the higher energy.

[3]

(c) The following table lists the colours and energies of photons of light of certain wavelengths.

wavelength /nm	energy of photon	colour of photon
400	high	violet
450	↓	blue
500	lower	green
600	↓	yellow
650	low	red

The visible spectra of solutions of two transition metal complexes **C** and **D** are shown in the diagram below.



(i) A list of possible colours for these complexes is as follows.

yellow red green blue

Choose one of these words to describe the observed colour of each solution.

(iii) In which complex, **C** or **D**, will the energy gap between the two groups of orbitals be the larger? Explain your answer.

[3]

[Total 8 marks]

[© UCLES 2014 9701/4/M/J/07]

23. The ions of transition elements form *complexes* by reacting with *ligands*.

(a) (i) State what is meant by the terms:

1. *complex*;

2. *ligand*.

(ii) Two of the complexes formed by copper are $[Cu(H_2O)_6]^{2+}$ and $CuCl_4^{2-}$.

Draw three-dimensional diagrams of their structures and name their shapes.

(iii) Platinum forms square-planar complexes, in which all four ligands lie in the same plane as the Pt atom.

There are two isomeric complexes with the formula $Pt(NH_3)_2Cl_2$.

Suggest the structures of the two isomers, and, by comparison with a similar type of isomerism in organic chemistry, suggest the type of isomerism shown here.

[7]

(b) Copper forms two series of compounds, one containing copper (II) ions and the other containing copper (I) ions.

(i) Complete the electronic structures of these ions.

(ii) Use these electronic structures to explain why copper (II) salts are usually coloured, while copper (I) salts are usually white or colourless.

[5]

[Total 12 marks]

[© UCLES 2014 9701/41/M/J/14]

23. (a) Describe and explain what would be observed when aqueous ammonia is added gradually to cobalt (II) chloride until it is present in excess and the product is allowed to stand in air.

[4]

(b) **A** and **B** are compounds of formula $CrCl_3.6H_2O$. an aqueous solution containing $0.002\ moles$ of **A** required $25.0\ cm^3$ of $0.080\ mol\ dm^{-3}$ aqueous $AgNO_3$ and **B** required $50.0\ cm^3$ of $0.080\ mol\ dm^{-3}$ aqueous $AgNO_3$.

(i) Calculate the number of moles of chloride ions present in one mole of

1. compound **A;**

2. compound **B.**

(ii) Suggest structural formulae for compounds **A** and **B.**

(iii) Draw diagrams to show isomers of **A.**

[6]

[Total 10 marks]

[© ZIMSEC 9189/4 J2011]

24. (a) (i) Give any **two** uses of cobalt.

(ii) Write the electronic configuration of Co^{3+}.

(iv) Explain why Co^{3+} ion is coloured.

[6]

(b) The pink hexaaqua cobalt (II) complex changes to a blue complex when concentrated HCl is added.

(i) Write a chemical equation for the reaction.

(ii) Draw the shape of the

1. pink complex

2. blue complex

[4]

[Total 10 marks]

[© ZIMSEC 9189/4 J2015**]**

25. The concentration of hydrogen peroxide in a solution can be determined by titrating it against an acidified solution of aqueous potassium manganate (VII). 250 cm^3 of a dilute aqueous solution of hydrogen peroxide were found to react with 18.10 cm^3 of 0.20 $mol\ dm^{-3}$ potassium manganate (VII) solution.

(a) Using suitable E^{θ} values from the Data Booklet write an equation for the reaction between acidified potassium manganate (VII) with hydrogen peroxide.

(b) State the role of hydrogen peroxide in the reaction.

(c) Identify with a reason a suitable acid for this reaction.

(d) State the colour changes at the end-point of the titration.

(e) Calculate the molar concentration of hydrogen peroxide in the reaction.

[Total 10 marks]

[© ZIMSEC 9189/4 N2006**]**

26. All five d-orbitals in the outer d sub-shell of an isolated transition element ion have the same energy. When ligands bond to the central ion, the d-orbitals move to two different energy levels. The gap between the energy levels depends upon

I the transition metal ion;

II the ligand;

III the coordination.

(a) What do you understand by the term *ligand*?

[1]

(b) Using copper as an example discuss how the **three** factors determine the colour in its ions and complexes.

[9]

[Total 10 marks]

[© ZIMSEC 9189/4 N2006**]**

27. (a) Define the terms

1. *ligand*

2. *dative bond*, and

3. *complex ion.*

[3]

(b) Ethane-1,2-diaminetetraacetate ion, $EDTA^{4-}$, is a polydentate ligand.

Draw the structure of $EDTA^{4-}$ and indicate all the centres of dative bond formation in the ion.

[3]

(c) Describe and explain what will be observed when aqueous ammonia is added to a solution containing Cu^{2+} ions until it is in excess, followed by the addition of $EDTA$.

You may use data in **Table 2.**

Table 2

complex ion	stability constant
$[Cu(NH_3)_4]^{2+}$	$\mathbf{1.4\ x\ 10^{13}}$
$[Cu(EDTA)]^{2-}$	$\mathbf{6.8\ x\ 10^{18}}$

[4]

[Total 10 marks]

[© ZIMSEC 9189/4 N2012**]**

28. (a) Describe, with the aid of one suitable example in each case, the commonly occurring shapes of complex ions formed by transition metals.

[4]

(b) Two isomeric complex ions of chromium (III) exist which have the formula $[Cr(C_2O_4)_3]^{3-}$.

(i) What type of ligand is $C_2O_4^{2-}$?

(ii) Draw **two** structural formulae of these isomeric ions and name the form of isomerism shown by these isomers.

(iii) Draw another type of isomerism shown by octahedral complexes other than the one above.

[9]

[Total 10 marks]

29. (a) Complexes with polydentate and bidentate ligands have larger stability constants than those of monodentate ligands.

(i) Define the terms

1. bidentate;

2. polydentate.

(ii) Using $[M(H_2O)_6]^{n+}$ as an example, explain the term *stability constant.*

[6]

(b) **Table 1** shows stability constants of some copper complexes.

Table 1

ion with water as ligand	complex	Log(stability constant)
Cu^{2+}	$[Cu(NH_3)_4]^{2+}$	**13.1**
Cu^{2+}	$[CuCl_4]^{2-}$	**5.6**

With reference to **Table 1** suggest what would happen if small amounts of concentrated ammonia are added to $[CuCl_4]^{2-}$ solution until present in excess.

[4]

[Total 10 marks]